APPLICATION FOR PATENT

FORM 2 — COMPLETE SPECIFICATION

# COMPACT STM32-BASED EMBEDDED SYSTEM BOARD WITH INTEGRATED LI-ION BATTERY MANAGEMENT AND HIGH-EFFICIENCY DC-DC POWER CONVERSION

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## 1. FIELD OF THE INVENTION

The present invention relates to embedded systems hardware design, and more particularly to a compact, single-board microcontroller system integrating an STM32 microcontroller unit with an onboard lithium-ion battery charging circuit, a high-efficiency DC-DC buck-boost converter, and a USB-C interface — all implemented on a 2-layer printed circuit board (PCB) suitable for real-world product deployment.

## 2. BACKGROUND OF THE INVENTION

Conventional STM32 development kits such as the Nucleo and Discovery series are widely used for rapid prototyping of embedded systems. However, these boards exhibit critical limitations when transitioning from evaluation to product deployment:

1. Bulky form factors that are incompatible with compact enclosures.
1. Absence of onboard lithium-ion battery management, requiring separate charger modules such as the TP4056.
2. Reliance on linear voltage regulators (LDOs) that exhibit poor power conversion efficiency.
3. Requirement for external boost/buck converter modules, increasing wiring complexity and overall system footprint.
4. No support for autonomous battery-powered operation without significant hardware additions.

There exists a need for a compact, self-contained embedded board that addresses these limitations by integrating all power management, charging, and processing functionalities onto a single, optimized PCB.

## 3. SUMMARY OF THE INVENTION

The present invention provides a compact single-board embedded system comprising an STM32 microcontroller, an integrated lithium-ion battery charging circuit, a high-efficiency DC-DC power conversion stage, a USB-C connector for power delivery and programming, and an SWD debug interface — all implemented on an optimized 2-layer PCB. The invention eliminates the need for external power management modules, reduces overall system size, and enables autonomous battery-powered operation suitable for IoT devices, portable instruments, and industrial monitoring applications.

# 4. DETAILED DESCRIPTION OF THE INVENTION

## 4.1 Overall Architecture

The invention is a single printed circuit board integrating the following functional blocks:

- STM32 Microcontroller Unit (MCU) — the primary processing element.
- USB-C Power and Data Interface — provides 5V input, USB data lines (D+/D−), and CC pin resistors for cable orientation detection.
- USB-UART Bridge (CH340 or equivalent) — enables serial communication and firmware upload without a dedicated programmer.
- Li-Ion Battery Management IC (TPS63802DLAR) — provides simultaneous charging and power path management for a single-cell lithium-ion battery.
- DC-DC Buck-Boost Converter — efficiently converts battery voltage to a regulated 3.3V MCU supply rail, maintaining regulation across the full battery discharge range.
- SWD Debug Interface — standard 2-wire Serial Wire Debug port for firmware debugging and programming.

## 4.2 Power Management Subsystem

The power management subsystem is a key novelty of the invention. Unlike conventional development boards that rely on a simple LDO regulator chain, the present invention employs a DC-DC buck-boost topology. This allows the board to operate efficiently across the full lithium-ion discharge curve (4.2V to 2.8V), delivering a stable 3.3V to the MCU at all times. The TPS63802DLAR IC manages both the charging of the battery from the USB-C VBUS rail and the regulated output to the system, eliminating the need for discrete charging and regulation components.

## 4.3 USB-C Interface and USB-UART Bridge

The USB-C connector serves dual roles: power delivery (VBUS at 5V) and data communication. The CC1 and CC2 pins are pulled down via resistors (R13, R14 at 5.1kΩ) to comply with USB-C specifications for device-side power sink configuration. The D+ and D− lines are routed through series resistors (R17, R18 at 22Ω) to the USB-UART bridge IC, which provides USART1_TX and USART1_RX connections to the STM32 for serial firmware upload and debug output.

## 4.4 PCB Design Considerations

The entire system is implemented on a 2-layer PCB to minimize manufacturing cost while maintaining signal integrity. Decoupling capacitors (100nF and 4.7μF) are placed at all power supply pins. The compact layout is designed to fit within space-constrained enclosures, making it suitable for product deployment in IoT nodes, wearables, portable instruments, and remote industrial sensors.

## 5. CLAIMS

**We claim:**

5. A compact embedded system board comprising: an STM32 microcontroller unit; a USB-C interface connector providing power and data connectivity; a USB-UART bridge circuit connected to the USART peripheral of the STM32 microcontroller; a lithium-ion battery management integrated circuit providing simultaneous battery charging and power path control; a DC-DC buck-boost converter providing a regulated 3.3V supply from the battery output; and all of the above implemented on a single 2-layer printed circuit board.
6. The board of claim 1, wherein the USB-C interface includes CC pin pull-down resistors for USB Power Delivery sink configuration and series resistors on D+ and D− lines for USB signal integrity.
7. The board of claim 1, wherein the DC-DC converter maintains regulated output voltage across a lithium-ion battery discharge range of 2.8V to 4.2V.
8. The board of claim 1, wherein the board includes an SWD (Serial Wire Debug) interface for firmware programming and debugging.
9. The board of claim 1, wherein no external power management modules, battery charger modules, or boost/buck converter modules are required for autonomous battery-powered operation.

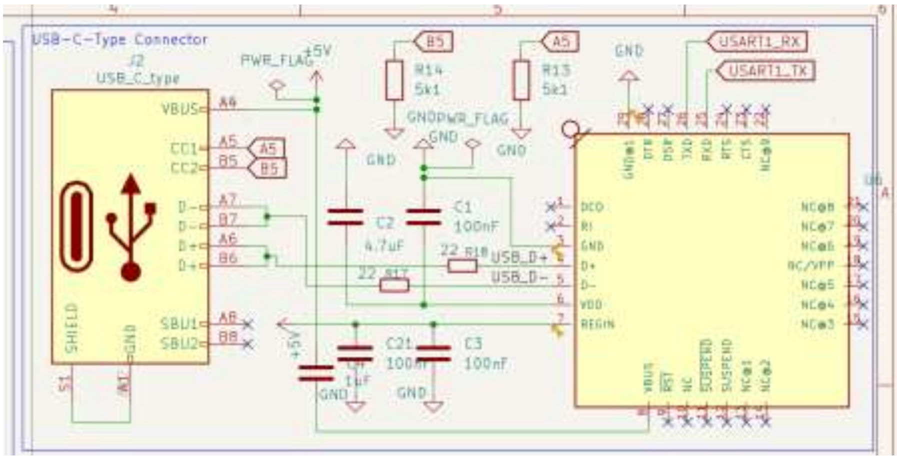
## 6. ABSTRACT

A compact embedded system board is disclosed, integrating an STM32 microcontroller, onboard lithium-ion battery management, a DC-DC buck-boost power converter, a USB-C interface, and a USB-UART bridge on a single 2-layer PCB. The invention overcomes the limitations of conventional development kits — which require external battery charger modules, boost/buck converters, and have bulky form factors — by providing an all-in-one, product-ready embedded platform. The board is suitable for IoT devices, portable instruments, wearables, and remote industrial monitoring systems requiring autonomous, battery-powered operation.

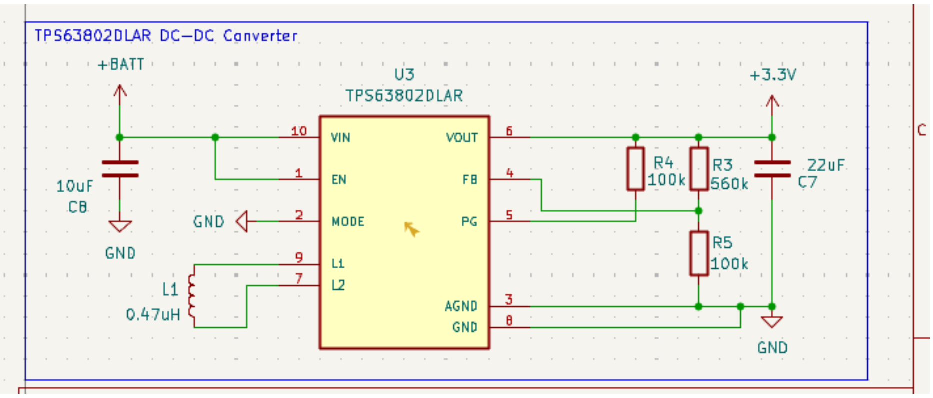
## 7. DRAWINGS

The following drawings form part of the present specification and are included to further demonstrate certain aspects of the invention.

## Figure 1 — USB-C Interface and USB-UART Bridge Schematic




## Figure 2 — Power Management Subsystem Schematic




## Figure 3 — Complete PCB Layout (Top and Bottom Layer)

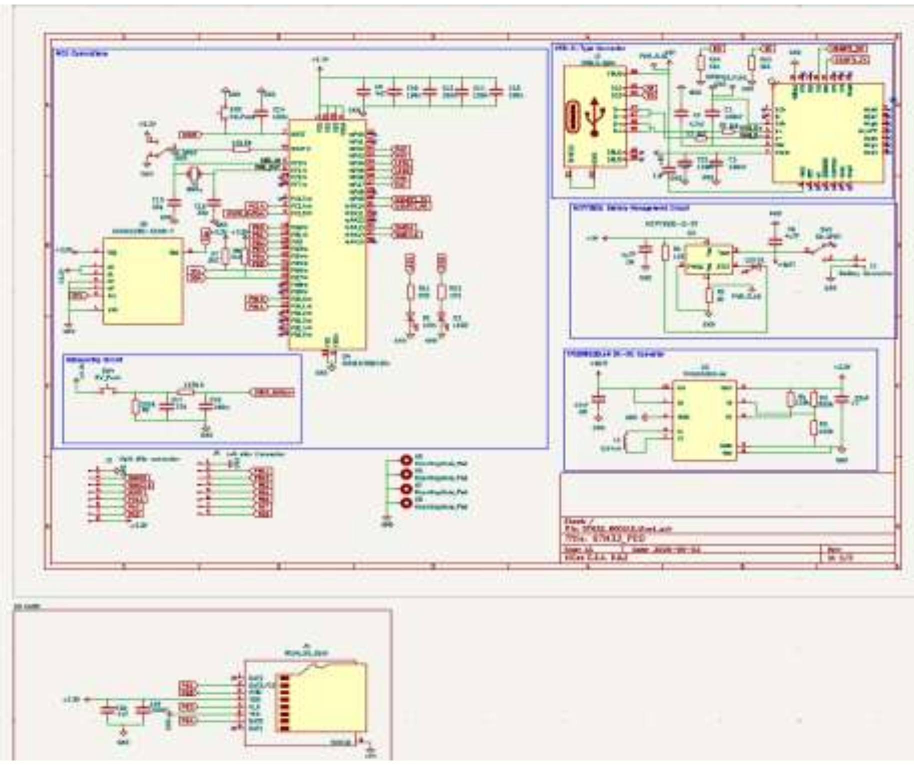
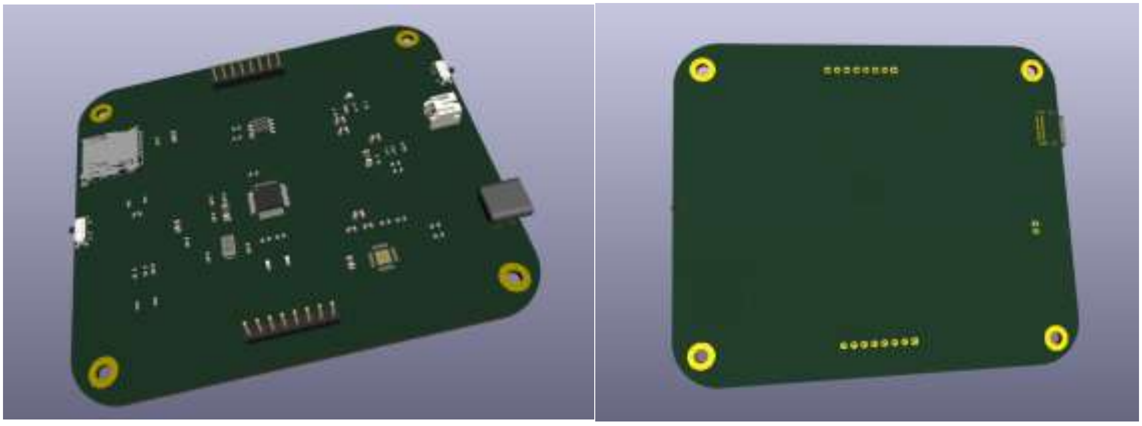


**Figure 4 — 3D Diagram of the product (without wiring )**



## 8. DECLARATION

I, Abhinandh K (Registration No: 192312276), hereby declare that I am the true and first inventor of the invention described in this specification, and that the invention has not been disclosed publicly prior to this filing.

Signature: ______________________

Name: Abhinandh K

Registration No: 192312276

Date: ______________________